ALPACA AI TRADING AGENTS HACKATHON · TEAM QWERTYS · AUG 28 – SEP 4, 2026

Θ THETA DESK

An autonomous options desk that prices **volatility**, not direction.

18
deterministic risk gates — LLMs cannot loosen a single one

4
LLM roles that argue, veto and size — Claude + open models via Featherless

100%
of decisions replay bit-for-bit from a hash-chained journal

4
books running live: real + no-gates + no-hedge + naive baseline

Alpaca Trading API · MCP Server · CLI — paper account PA39C10YAMYQ

github.com/pip00sya/theta-desk

THE THESIS

Over five days, price prediction is a coin flip. The volatility risk premium is measurable.

What everyone builds

«LLM reads headlines → buys calls.» A directional bet over 5 days: ~50% win rate, minus spread, minus theta. Negative expected value with extra steps.

What THETA DESK does

Compares ATM implied volatility (Alpaca chain, free feed) against 20-day realized volatility — a deterministic VRP score. Rich vol → sell iron condors, harvest theta. Cheap vol → buy convexity, micro-size.

Day one proof

The plan said «sell condors». The live signal said RV 13.9% > IV 11.6% — premium was cheap. The agent **went against our own plan** and bought puts instead. The system follows the signal, not the narrative.

Regime map: rich → condors · neutral → wider, half size · cheap → micro long vega · QQQ fallback

ARCHITECTURE — ONE TICK, EVERY 15 MINUTES OF THE SESSION

LLMs decide whether it's wise.
Code decides whether it's allowed.

Data
chain + greeks + IV (free feed), 20d RV, positions, per-underlying spots

→

Desk · 4 LLM roles
Vol Analyst (Claude) · Second Opinion (Mistral) · News Vetoer (Qwen) · Risk Officer (Claude). Disagreement → ×0.5 size. Veto → no new risk

→

Selector
deterministic regime map picks the structure and delta-targets the strikes

→

18 gates
incl. ★ portfolio payoff simulator — the desk's veto right

→

Execution
mleg limit via Alpaca CLI, signed prices, idempotent order ids

→

Audit
hash-chained journal · snapshots · 4 books marked

Management, every tick

35% target on credit · +60% target on debit · realization policy on idle days · NFP de-risk locks winners ≥ +15% the day before · drawdown → halt, never panic-flatten

Sizing is earned

Risk budgets grow only from **realized** gains: worst-case 6% → 8%, long-premium sleeve 1.5% → 2.5%. The agent earns the right to take risk

★ THE VETO RIGHT

Before any order, the whole book is repriced.

Every open leg plus the candidate, over a ±20% underlying-price grid at the judging horizon (deadline + 14 days — nothing expires while judges look), under a base and a stressed-vol scenario, each leg priced off its own underlying's spot. If the worst grid P&L breaches budget — the order is never sent, and the refusal is journaled with the full grid.

Why it matters

This is a client-side implementation of the same worst-case principle as **Alpaca's universal spread rule** for options margin — applied one step earlier in the pipeline. We refuse orders before the broker ever could.

Refusals are the product

An agent that can decline to trade is the point. Every refusal is a first-class journal entry: which gate, why, at what worst-case number. The dashboard shows them next to the fills.

Tested like an engine

58 unit + property tests: condor worst case = one wing; hedge converts the tail, not the plateau; random books never pass beyond budget; missing spot raises instead of silently mispricing.

engine/payoff.py · engine/gates.py — pure Python, no LLM anywhere near it

TRUST, MECHANICALLY

Every number regenerates. Every decision replays.

Hash-chained journal

`entry_hash = sha256(ts + kind + data + prev_hash)`

Tamper-evident by construction. `verify-journal` proves the chain; a single edited byte breaks it.

Bit-for-bit replay

Every tick stores its full input snapshot. `tools/replay.py` re-runs the deterministic pipeline over the week and diffs each decision. Current score: **100% MATCH**. Claims: **20/20**, zero mismatches.

Claims reconciler

`tools/reconcile.py` recomputes every number in the write-up from the journal — no credentials required. Evidence archive snapshots account/orders/positions daily at the close.

DEVLOG — 33 documented self-corrections

The build found real things: Alpaca's mleg **limit price is signed** (negative = credit — our abs() once turned «collect ≥ 3.40» into «pay ≤ 3.40», caught by a weekend acceptance test, fixed, upstream feedback written) · paper fills options against Friday quotes while closed · battery defaults silently skip scheduled runs · a far-OTM hedge converts the tail, not the worst case. Each one is in the repo with its fix and test.

make verify = chain + replay + reconcile, one command, green

LIVE ABLATION — THE DOLLAR VALUE OF EVERY DESIGN DECISION

Four books. Same inputs. Different brains.

BOOK	WHAT IT IS	P&L (UPDATED AT SUBMISSION)
real	the full agent: desk + 18 gates + management	**+$763**
shadow_nogates	same strategy, gates ignored, fills at mid	**+$61** — the gates are worth **$702**
shadow_nohedge	the real book minus its hedge sleeve	**+$772** — the tail cost **$9** so far
baseline_naive	«LLM reads a headline, buys ATM» — what a coin flip earns	**−$46** — negative all week

We didn't just claim the gates and the structure matter — we ran the agent without them, on the same ticks, and published the gap.

Marks carry net book greeks ($-delta / $-theta / $-vega) — the dashboard explains WHY the curve moved

Small, explained, and reproducible P&L beats a lucky coin flip.

The week in one story

Day one: vol was cheap by our own measure — the agent bought convexity instead of selling premium. The market slid; three of four puts closed at **+64…70%** for **+$764** realized. Then realized vol collapsed to 7.2% against 12.4% implied, the regime flipped **rich**, and the desk sold its first iron condors — the structure it was built for. Both branches, live, on their own signal.

Risk that never left the box

Max structural loss per position ≤ 1.25% equity · book worst case gate-capped at 7.5–8% · long-premium sleeve ≤ 1.5–2.5% (earned) · NFP-morning deadline entered with winners locked by rule.

Built and operated by AI

Designed, coded, red-teamed, debugged and run end-to-end by Claude (Claude Code + Alpaca MCP), with 33 self-corrections documented in DEVLOG.md — including the desk replaying its own snapshot to find a **$208** mispriced exit, then pinning it with a test. Disclosed openly.